

EXERCISES AND EXACT SOLUTIONS

V7_B_N12 - Health Surveys and Small-Area Evidence

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C32
Notebook	V7_B_N12
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Compare weighted and unweighted prevalence.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add effective sample size.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Explain model-assisted release safeguards.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Weighting represents unequal selection and response; disagreement is a diagnostic, not a reason to discard weights.

Solution 2

Use squared-weight concentration as a simple diagnostic and the full design for official variance.

Solution 3

Pre-specify models, validate domains, publish uncertainty, protect confidentiality, and retain official release review.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated